Certified Analytical Bounds for Multi-Phase Hypersonic Reachable Sets

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Abstract

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Contents

Open items

1 Introduction

1.1 Grid-based Hamilton-Jacobi-Isaacs evaluations

1.2 Fixed-grid pseudospectral methods

1.3 Thesis statement

1.4 Outline of contributions

2 The Model Problem

2.1 Abstract formulation of the multi-phase optimal control problem

The vehicle is a rigid body, not a point mass (Appendix A). Its state

$$x = \big(r, \theta, \phi,\ u, v, w,\ q_0, q_1, q_2, q_3,\ \omega_1, \omega_2, \omega_3\big) \in \mathcal{X} \subseteq \mathbb{R}^{13}, \tag{2.1}$$

augmented by mass, recession depth s and modal state $(\eta, \dot\eta)$ where the corresponding mode or coupling is active, carries geocentric position, body-frame velocity, attitude quaternion and body rates as components. The attitude coordinates are redundant: every admissible state satisfies

$$q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1, \tag{2.2}$$

so the thirteen coordinates carry twelve degrees of freedom. The control

$$c = \big(\vec\delta,\ \vec\tau_{\mathrm{rcs}},\ \varsigma\big) \in U, \tag{2.3}$$

collects flap deflections $\vec\delta$, reaction-control moments $\vec\tau_{\mathrm{rcs}}$ and, in a propulsive mode, throttle ς; $U \subset \mathbb{R}^{n_c}$ is compact.

list deflection, RCS, and throttle limits (one citation each). compactness follows.

Crucially the incidence angles (α, β) are *outputs* of the attitude solution (Eq. (A.5)), not commanded quantities: trim is not assumed solvable, and attitudes the vehicle cannot hold are excluded automatically rather than by fiat. The reachable set cannot contain a state reached only by an unattainable incidence.

elaborate

2.2 The mode alphabet

2.3 Trajectory families as admissible words

2.4 The entry corridor: hypersonic path constraints

2.5 Certified envelopes for constitutive and coupling effects

2.6 Phase boundaries: physical, constitutive, and numerical

zhao2020optimal asm. 2 (p2063): the field has to be transverse to every guard. that constrains where the $\bar q$ threshold can sit (a trajectory grazing the surface tangentially breaks it, prove transversality for the corridor? restrict the threshold placement like altitude band?).

2.7 Asymptotic regimes and the dimensionless groups

2.8 Quantity of interest and uncertainty

3 Infinite-Dimensional Embedding and Measure Theory

3.1 Sobolev space formulation

3.2 Occupation measures and the weak Liouville identity

the converse comes with a hypothesis attached. zhao2020optimal asm. 11 (p2066): need $F(t, x, U)$ compact *and convex* or measures satisfying Liouville need not come from trajectories. cor. 13 (p2066) gives control-affine dynamics with convex U as sufficient (flap deflections enter nonlinearly? check.)

3.3 Per-mode measures, guard measures, and linkage

3.4 The embedding as a linear problem on a Banach space

4 Concentration-Compactness and Profile Decomposition

4.1 Prokhorov tightness as a preliminary

4.2 The singular limit and the scaling group

4.3 Tightness of the rescaled family

4.4 The dichotomy

4.5 The skeleton: concentration on the equilibrium glide manifold

4.6 The bubbles: boundary-layer profiles at each skip

4.7 The compact residual

4.8 Consequence: dimension reduction for the certificate

5 Bounding Limits and Certificates

5.1 Relaxation and the direction of inclusion

5.2 The comparison principle

5.3 Constructing the certificate from the decomposition

5.4 Verification over the corridor

5.5 Exact arithmetic and machine-checkable certification

5.6 Inner point cloud and the bounding gap

5.7 Limits of the analytical construction

6 Verification and Results

6.1 Validating the degenerate limit

6.2 Tightness against the extremal cloud

6.3 Independent numerical cross-check

6.4 The corridor constraints as hypotheses

6.5 Multi-mode words

6.6 Numerical evidence for the profile decomposition

6.7 Computational cost

6.8 Reproducibility environment

7 Discussion

7.1 Limitations

7.2 Scope: the boost-glide and aeroassisted class

7.3 Extensions

References

A Six-Degree-of-Freedom Equations of Motion

This appendix records the rigid-body dynamics the body of the paper treats as given. Nothing here is a contribution; the derivation is standard entry mechanics assembled once with a fixed set of frames and a fixed parametrization. Frame errors are the classic 6-DOF defect and are invisible in the assembled algebra, so the transformations are stated explicitly and used without exception.

cite sources

A.1 State vector and frames

Frames. Four frames are used throughout.

- $\mathcal{I}$: Earth-centred inertial (ECI), non-rotating.
- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E\, \hat{z}_{\mathcal{I}}$.
- $\mathcal{L}$: local geocentric frame at the vehicle, with unit vectors $(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi)$ pointing radially outward, east, and north.
- $\mathcal{B}$: body frame, fixed at the instantaneous centre of mass, axes along the vehicle's principal design directions.

The attitude quaternion $\vec{q}$ carries $\mathcal{I} \to \mathcal{B}$; the geocentric angles (θ, ϕ) carry $\mathcal{I} \to \mathcal{L}$.

The direction-cosine matrix built from $\vec{q}$,

$$\mathbf{R}_{B/I}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 - q_0q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \tag{A.1}$$

is polynomial of degree two in $\vec{q}$, which is the property Appendix C exploits. The geocentric map is

$$\mathbf{R}_{L/I}(\theta, \phi) = \begin{pmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{pmatrix}. \tag{A.2}$$

State vector. Before augmentation the state carries thirteen components,

$$x = \big(\underbrace{r, \theta, \phi}_{\text{position}}, \underbrace{u, v, w}_{\text{body velocity}}, \underbrace{q_0, q_1, q_2, q_3}_{\text{attitude}}, \underbrace{p, q, r_b}_{\text{body rates}}\big) \in \mathbb{R}^{13}, \tag{A.3}$$

augmented by mass m when a propulsive mode is present (Appendix A.6), and by the recession depth s and modal state $(\eta, \dot{\eta})$ once the aerothermoelastic loop is closed (Appendix A.3 and Section 2.5). Here (r, θ, ϕ) are geocentric radius, longitude and latitude; (u, v, w) are the body-axis components of the *planet-relative* velocity

$$\vec{V} \;=\; \vec{V}_{\mathcal{I}} - \vec{\omega}_E \times \vec{r} - \vec{w}_{\text{wind}}, \tag{A.4}$$

with $\vec{V}_{\mathcal{I}}$ the inertial velocity and $\vec{w}_{\text{wind}}$ the local wind (Appendix A.5); and (p, q, r_b) are the body components of the inertial angular velocity $\vec{\omega} \equiv \vec{\omega}_{B/I}$.

Remark A.1 (Why body-frame velocity, not (V, γ, ψ)). Writing velocity in body axes makes the incidence angles ratios of state components rather than inverse-trigonometric functions of them:

$$\alpha = \operatorname{atan2}(w, u), \qquad \beta = \arcsin\big(v/V\big), \qquad V = \sqrt{u^2 + v^2 + w^2}. \tag{A.5}$$

Adjoining V through the polynomial constraint $V^2 = u^2 + v^2 + w^2$ lets the aerodynamic coefficients be written in the direction cosines $(u/V,\ v/V,\ w/V)$, which is what carries them into the polynomial ring of Appendix C.

A.2 Translational equations

Newton's law holds in $\mathcal{I}$: $m\,\dot{\vec{V}}_{\mathcal{I}}\big|_{\mathcal{I}} = \vec{F}$, with $\vec{F} = \vec{F}_g + \vec{F}_a + \vec{F}_t$ (gravity, aerodynamic, thrust). We want the rate of the *relative* velocity resolved in $\mathcal{B}$. Writing $\vec{V}_{\mathcal{I}} = \vec{V} + \vec{\omega}_E \times \vec{r}$ and differentiating in $\mathcal{I}$,

$$\dot{\vec{V}}_{\mathcal{I}}\big|_{\mathcal{I}} = \dot{\vec{V}}\big|_{\mathcal{I}} + \vec{\omega}_E \times \vec{V}_{\mathcal{I}} = \underbrace{\dot{\vec{V}}\big|_{\mathcal{B}} + \vec{\omega} \times \vec{V}}_{\text{transport of } \vec{V}} + \vec{\omega}_E \times \vec{V} + \vec{\omega}_E \times (\vec{\omega}_E \times \vec{r}), \tag{A.6}$$

where the transport theorem $\frac{d}{dt}\big|_{\mathcal{I}} = \frac{d}{dt}\big|_{\mathcal{B}} + \vec{\omega}\times$ was applied to $\vec{V}$. Solving for the body-frame rate,

$$\boxed{\dot{\vec{V}}\big|_{\mathcal{B}} = \frac{1}{m}\big(\vec{F}_a + \vec{F}_t\big) + \vec{g} - \big(\vec{\omega} + \vec{\omega}_E\big) \times \vec{V} - \vec{\omega}_E \times (\vec{\omega}_E \times \vec{r})} \tag{A.7}$$

with $\vec{F}_g = m\vec{g}$. The term $(\vec{\omega} + \vec{\omega}_E) \times \vec{V}$ collects body-rotation transport and Coriolis; expanding $\vec{\omega} = \vec{\omega}_{B/E} + \vec{\omega}_E$ recovers the familiar $\vec{\omega}_{B/E} \times \vec{V} + 2\,\vec{\omega}_E \times \vec{V}$ split, the second being the Coriolis acceleration proper.

Resolved in $\mathcal{B}$ with $\vec{V} = (u, v, w)^\top$, $\vec{\omega} = (p, q, r_b)^\top$, $\vec{\omega}_E^{\mathcal{B}} = \mathbf{R}_{B/I}\,\omega_E \hat{z}_{\mathcal{I}}$, and $\vec{r}^{\mathcal{B}} = \mathbf{R}_{B/I}\, r\,\hat{e}_r$, Eq. (A.7) is three scalar ODEs for $(\dot{u}, \dot{v}, \dot{w})$; the full expansion is assembled in `src/aether/dynamics/`.

Gravity. The central term is $\vec{g} = -(\mu/r^2)\,\hat{e}_r$. On long vacuum arcs the oblateness perturbation is retained; in geocentric components,

$$g_r = -\frac{\mu}{r^2}\Big[1 - \tfrac{3}{2}J_2\big(R_e/r\big)^2\big(3\sin^2\phi - 1\big)\Big], \tag{A.8}$$

$$g_\phi = -\frac{\mu}{r^2}\,3J_2\big(R_e/r\big)^2 \sin\phi\cos\phi, \tag{A.9}$$

with $g_\theta = 0$, then rotated into $\mathcal{B}$ by $\mathbf{R}_{B/I}\mathbf{R}_{L/I}^\top$.

Keeping J2 on vacuum coasts and dropping inside the atmospheric mode, where aerodynamic acceleration dominates the oblateness term by several orders of magnitude over the corridor. Stating this explicitly or carry J2 throughout.

Position kinematics. With $(V_r, V_\theta, V_\phi)^\top = \mathbf{R}_{L/I}\mathbf{R}_{B/I}^\top(u, v, w)^\top$ the relative velocity in local geocentric axes,

$$\dot{r} = V_r, \qquad \dot{\theta} = \frac{V_\theta}{r\cos\phi}, \qquad \dot{\phi} = \frac{V_\phi}{r}. \tag{A.10}$$

A.3 Rotational equations

Angular momentum about the center of mass is $\vec{h} = \mathbf{I}\vec{\omega}$, and $\dot{\vec{h}}\big|_{\mathcal{I}} = \vec{M}$; transporting to $\mathcal{B}$,

$$\boxed{\mathbf{I}\,\dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I}\vec{\omega}) - \dot{\mathbf{I}}\,\vec{\omega}} \tag{A.11}$$

with $\vec{M} = \vec{M}_a + \vec{M}_{\text{rcs}}$. The gyroscopic term is quadratic in $\vec{\omega}$, hence polynomial once $\mathbf{I}$ is; the inversion $\dot{\vec{\omega}} = \mathbf{I}^{-1}(\cdots)$ is carried as the linear solve $\mathbf{I}\dot{\vec{\omega}} = (\cdots)$ so that no rational function of the state enters the embedding.

Remark A.2 (The inertia tensor is not constant). Ablation removes mass from the nose and structural deformation redistributes it, so $\mathbf{I} = \mathbf{I}(s, \eta)$ through the recession depth s and modal amplitudes η (Section 2.5), and Euler's equations acquire the $\dot{\mathbf{I}}\vec{\omega}$ term in Eq. (A.11). We carry it rather than drop it: the whole point of maintaining 6-DOF is to keep the deformation→incidence→moment→attitude loop closed, and that loop runs through $\mathbf{I}(s, \eta)$. Its magnitude is bounded by the recession and modal-amplitude envelopes, so carrying it costs one Jacobian block, not a new uncertainty.

Remark A.3 (Variable-mass reaction terms). Mass leaving by ablation carries angular momentum with it. Modelling the char products as departing at the local surface velocity (no net ablative thrust), the reaction torque is second order in the recession rate and is absorbed into the moment envelope of Appendix A.4; only the $\dot{\mathbf{I}}\vec{\omega}$ contribution is kept explicitly. Propulsive mass flow enters through $\vec{M}_{\mathrm{rcs}}$ and the mass equation, not here.

Attitude kinematics. With $\vec{q}$ carrying $\mathcal{I} \to \mathcal{B}$ and body rates $\vec{\omega} = (p, q, r_b)$,

$$\dot{\vec{q}} = \tfrac{1}{2}\,\Omega(\vec{\omega})\,\vec{q}, \qquad \Omega(\vec{\omega}) = \begin{pmatrix} 0 & -p & -q & -r_b \\ p & 0 & r_b & -q \\ q & -r_b & 0 & p \\ r_b & q & -p & 0 \end{pmatrix}, \tag{A.12}$$

bilinear in $(\vec{q}, \vec{\omega})$ and hence polynomial. The unit-norm condition $|\vec{q}|^2 = 1$ is a polynomial constraint the embedding already carries rather than one it must add; Ω is skew, so the flow preserves $|\vec{q}|$ exactly and the constraint is invariant, not merely initial.

A.4 Aerodynamic forces and moments

Six coefficients, with dynamic pressure $\bar{q} = \frac{1}{2}\rho V^2$, reference area S and reference length ℓ_{ref},

$$\vec{F}_a = \bar{q}\,S \begin{pmatrix} C_X \\ C_Y \\ C_Z \end{pmatrix}, \qquad \vec{M}_a = \bar{q}\,S\,\ell_{\mathrm{ref}} \begin{pmatrix} C_l \\ C_m \\ C_n \end{pmatrix}, \tag{A.13}$$

each coefficient a function of the incidence pair, the compressibility and rarefaction parameters, the control deflections, and (through Appendix B) the recession depth and modal amplitudes:

$$C_{(\cdot)} = C_{(\cdot)}\big(\alpha, \beta,\ M_\infty,\ Kn,\ \vec{\delta},\ s,\ \eta\big), \qquad M_\infty = \frac{V}{a(h)}, \quad Kn = \frac{\lambda(h)}{\ell_{\mathrm{ref}}}, \tag{A.14}$$

with $a(h)$ the speed of sound and $\lambda(h)$ the mean free path from the atmosphere model.

A.5 Atmosphere

Density, speed of sound, and mean free path are taken from US Standard 1976 below 86 km, MSIS above, not a bare exponential. The exponential idealization $\rho = \rho_0 e^{-h/H_s}$ enters only in Appendix C, as the profile whose scale-height ratio $\varepsilon_{\mathrm{atm}} = H_s/R_e$ organizes the decomposition of Section 4; the departure of the tabulated profile from it is bounded there and enters the residual, not the skeleton. Geometric altitude is $h = r - R_e(\phi)$. Winds enter through Eq. (A.4) as a relative-wind correction; this matters more in 6-DOF than in 3-DOF because sideslip β is now a state-dependent output rather than an assumed zero, so a lateral gust perturbs the moment balance, not merely the force.